

IMAGINE Decoding Challenge

Brain Activity Classification using Magnetoencephalography (MEG)

Track 1 Kaggle Competition | Project Report

1. Project Objective

The primary goal of this project is to predict which of **10 distinct object categories** a participant is viewing based on their MEG brain activity [cite: 3]. This involves decoding high-dimensional neural signals to identify visual patterns processed by the brain.

10

Target Classes

1-40 Hz

Frequency Range

~10%

Baseline Accuracy

2. Assessment Evolution: Baseline vs. Improved

We transitioned from a simplified model to a robust, deep architecture to handle the complexities of MEG data.

Assessment 1: Baseline

Model: Simple CNN [cite: 6, 423]

Technique: 2-layer Conv1D with Global Average Pooling [cite: 177, 184].

Result: High noise sensitivity and low validation accuracy (~10%) due to lack of preprocessing [cite: 7].

Improved Submission

Model: Deep CNN with Residual Blocks [cite: 193, 436]

Technique: Multi-stage temporal extraction with Batch Normalization and Dropout [cite: 8, 199].

Result: Significant performance gain via signal cleaning and advanced pooling [cite: 313, 437].

3. Data Preprocessing Methodology

The raw MEG data is extremely noisy. We applied three critical signal processing techniques to extract meaningful features:

- **Bandpass Filtering (1-40 Hz):** This keeps brain-relevant frequencies while removing slow drifts (<1Hz) and high-frequency muscle noise (>40Hz) [cite: 62, 64].
- **Baseline Correction:** Subtracting the pre-stimulus mean makes each trial start from a zero reference point, reducing inter-trial variance [cite: 67, 68].
- **Z-Score Normalization:** Scaling features per channel ensures the model isn't biased by high-amplitude sensors [cite: 74].

```
# Technique: Signal Isolation
epochs.filter(l_freq=1.0, h_freq=40.0)
epochs.apply_baseline(baseline=(None, 0))
```

4. Advanced Architecture: Deep CNN

The improved model uses a series of functional blocks designed for time-series decoding [cite: 198]:

Block	Operation	Purpose
Temporal Block	64 filters, 7-kernel Conv1D	Extract initial time-domain patterns [cite: 201].
Pattern Block	128 filters, 5-kernel Conv1D	Detect complex signal shapes [cite: 211].
Dual Pooling	Concatenate(Avg + Max)	Captures both mean trends and peak activations [cite: 231, 456].

5. Training Strategy & Optimization

To prevent overfitting and ensure convergence, we implemented intelligent callbacks [cite: 262]:

- **Early Stopping:** Terminates training if validation accuracy plateaus for 10 epochs [cite: 267, 269].
- **ReduceLROnPlateau:** Halves the learning rate if the loss gets stuck, allowing for finer weight updates [cite: 274, 276].
- **L2 Regularization:** Prevents weights from growing too large, promoting better generalization [cite: 196, 456].

6. Conclusion & Future Roadmap

The transition from a raw baseline to a preprocessed deep learning pipeline significantly improved the model's ability to decode human vision from brain waves. Next iterations will focus on capturing temporal dependencies [cite: 462].

Future Targets:

- **v3:** LSTM/BiLSTM for sequence modeling [cite: 465].
- **v4:** Attention/Transformers for time-dimension weighting [cite: 467].
- **v5:** Data augmentation via Gaussian noise injection [cite: 468].